Lakshman Shaminthan

Software Engineer — Full-Stack Developer

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Summary

Computer Science and Engineering graduate with hands-on experience as a full-stack engineer, having built features across React/Next.js frontends, Node.js and Python backends, and MongoDB during my time at Cogntix, where I developed an AI-powered interview scheduling system. Currently engaged as a freelance full-stack developer on LankaRoute, an AI trip-planning platform. A quick learner looking to bring this hands-on product-building experience to a software engineering role.

Education

National Institute of Technology, Rourkela — **September 2022 – May 2026**
Bachelor of Technology in Computer Science and Engineering (First Class) — *Rourkela, Odisha*

Jaffna Hindu College — **December 2020**
Secondary School (Science Stream) — *Jaffna, Sri Lanka*

Experience

Cogntix — **Jaffna, Sri Lanka**
Software Engineer Intern — *Jan 2026 – Jul 2026*

- Worked as a full-stack developer on **PACC**, an eBook-selling platform with separate **Publisher, Admin, and Reader portals**, contributing across frontend, backend, and database layers.
- **Frontend:** Developed responsive user interfaces using **React and Next.js**.
- **Backend:** Built and maintained backend services using **Node.js**, including **REST API integration** for internal and third-party services, and **payment gateway integration** for transaction processing.
- **Database:** Designed and managed application data using **MongoDB** for efficient storage and retrieval.

Research and Development Intern — *May 2025 – Jul 2025*

- Developed an **Automated Interview Scheduling System** to streamline candidate outreach and recruiter scheduling.
- Built backend services using **Python, FastAPI, and MongoDB** for email automation and scheduling workflows.
- Integrated **AWS SES, SNS, and S3** for sending emails, real-time notifications, and email storage.
- Connected **Google Calendar API** to fetch recruiter availability and manage scheduling conflicts.
- Utilized **AI APIs (NLP models)** to analyze candidate emails for sentiment, date, and time availability.

Freelance Experience

LankaRoute – AI Trip Planning Platform — **Sri Lanka**
Full-Stack Developer — *Jul 2026 – Present*

- Engaged as lead full-stack developer on a 3-person team to build an AI-powered trip-planning platform for Sri Lanka.
- Architected the full-stack system: **Next.js** frontend (chosen for SEO) and a serverless backend on **Node.js, SST, AWS Lambda, and DynamoDB**.
- Designed a multi-step, card-based itinerary wizard guiding users from interest selection to a personalized, AI-generated itinerary.
- Deployed a live prototype and directed the team's technical architecture and design system.

Factory Machine Maintenance Simulation — *Flutter, Firebase, Node.js, React, MongoDB*
Full-Stack Developer — *Feb 2025 – Apr 2025*

- Engaged to build a simulation app and web platform to manage machine failures and adjuster assignments using Mean Time to Failure (MTTF).
- Implemented real-time failure simulations and queue-based logic, optimizing adjuster efficiency and minimizing downtime.
- Designed a MongoDB schema to track machine status, failure history, and adjuster assignment records.
- Built a Flutter mobile client alongside the React web app, syncing data through a shared Node.js backend.

Projects

SONAR Rock vs Mine Prediction | *Python, Machine Learning* | GitHub — **Nov 2024 – Dec 2024**

- Built an end-to-end ML model classifying sonar signals as rocks or mines using Logistic Regression and Random Forest, achieving 76% test accuracy.
- Compared multiple classification algorithms and tuned model parameters to improve prediction reliability.
- Evaluated model performance using accuracy scores and confusion matrices to validate classification quality.

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C, C++

Frontend/Backend: React, Next.js, HTML, CSS, Tailwind CSS, Flutter, Node.js, FastAPI, Express

Databases & Cloud: MongoDB, MySQL, AWS (S3, SES, SNS), Firebase

Other: REST API Development, Postman, AI API Integration (OpenAI, Anthropic), Git, WordPress, MATLAB

Achievements/Certifications

- **Python for Beginners, University of Moratuwa** — Sep 2024
- **IELTS** – Overall Band Score 6 — May 2022

Extracurricular Activities

Gavel Club — **Feb 2022 – Sep 2022**
Member — *Uki Coding School*

- Participated in workshops on leadership development, public speaking, and effective communication.